MM226: Materials Informatics

Assignment 2 – Materials Data Analysis

Alloy System: Cu–Zn Alloy

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1. Objective

The intention behind this assignment is to create hands-on experience with the preprocessing, analysis, visualization, and assessment of the materials datasets produced through the application of large language models (LLMs). In the present work, synthetic datasets related to the Cu-Zn alloy system, also referred to as brass, have been produced through the application of several AI tools and evaluated on the basis of their physical plausibility, consistency, and realism.

The study adds to the trends and metallurgical understanding of the properties of Cu-Zn alloys, emphasizing the connections among composition, processing, microstructure, and alloy properties. By comparing the datasets provided by different AI tools, the assignment illustrates the capability of AI-generated materials information, as well as the limitations of such information for exploratory materials informatics.

2. Part 1: AI-Generated

2.1 AI Tools Used

Hypothetical datasets for an experimental study of the Cu-Zn alloy system were created employing the following AI techniques:

- ChatGPT
- Google Gemini
- Perplex
- Copilot

Each AI tool was asked to produce 70-75 independent data points, prompting the data set to collectively fall within the given range of 240-320 data points.

2.2 AI Prompt Used (Same Structure for All Tools)

"You are a materials scientist generating a hypothetical experimental dataset for the Cu–Zn alloy system. Generate independent experimental data points including chemical composition (Cu, Zn, minor alloying elements in wt%), heat treatment conditions (solutionizing temperature, aging temperature and time), mechanical testing temperature, microstructural parameters (grain size, phases present, phase volume fractions), and mechanical properties (yield strength, ultimate tensile strength, elastic modulus, hardness with scale, elongation, strain rate). Introduce realistic variability, some missing values, minor unit inconsistencies, and small physical inconsistencies. Do not explain the data. Output only the table."

The raw outputs were preserved in CSV format without manual modification, in accordance with assignment guidelines.

2.3 Dataset Description

Each data point in the AI-generated datasets contains the following information:

Composition

- Copper (Cu wt%)
- Zinc (Zn wt%)
- Minor alloying elements such as Pb or Sn (occasionally present)

Processing Conditions

- Solutionizing temperature (°C)
- Aging temperature (°C)
- Aging time (hours)

Mechanical Testing Conditions

- Test temperature (°C)

Microstructural Parameters

- Average grain size (µm)
- Phases present (α, β, or α+β)
- Phase volume fractions (%)

Mechanical Properties

- Yield strength (MPa)

- Ultimate tensile strength (MPa)
- Elastic modulus (GPa)
- Hardness value with scale (HV / HB / HRB)
- Elongation (%)
- Strain rate (s^{-1})

Each AI-generated set was treated individually, as is preferred.

3. Part 2: Data Processing and Exploratory Analysis

3.1 Data Cleaning

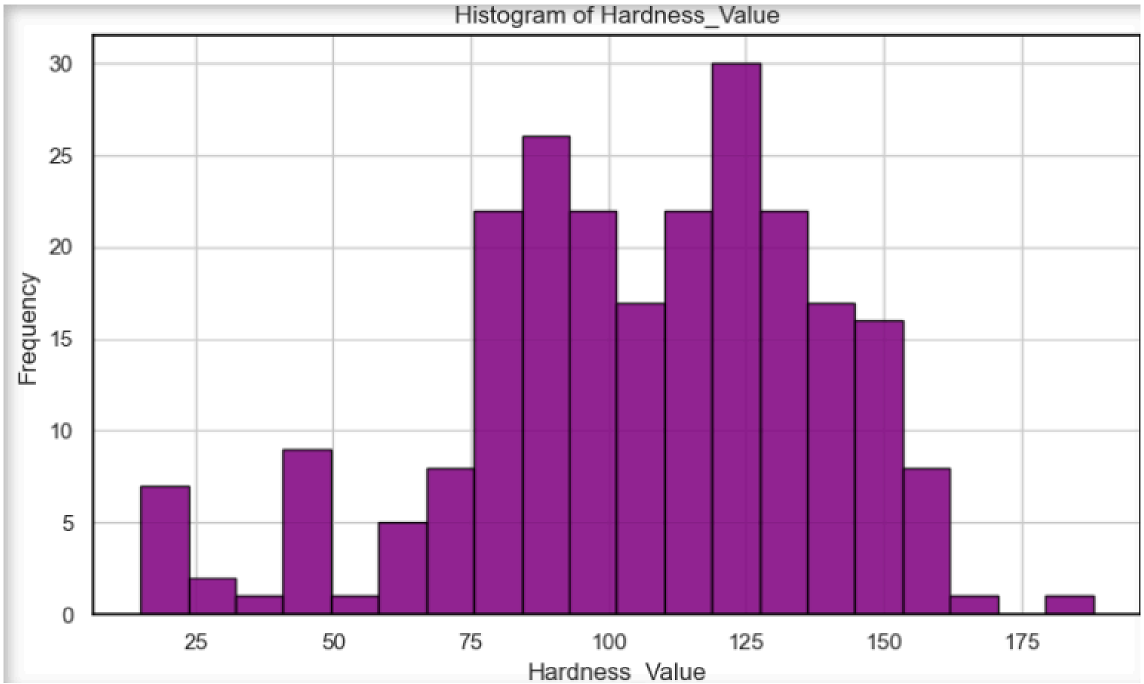
3.1.1 Column Harmonization

The varied naming conventions in the column headers from the different AI tools were inconsistent in naming the same physical variables, e.g., “YS”, “Yield_Strength”, “Yield Strength_MPa”. The naming of the data columns was standardized as follows before further analysis. This step mirrors the difficulties that have to be confronted in the practical application of multi-source materials databases.

3.1.2 Dealing

- Missing data was found in:
  - Minor alloying element content
  - Aging temperature or aging time
- The missing composition values have been replaced with 0.0 wt%, where Cu–Zn is considered to be a
- Missing processing parameters were kept as NaN and treated accordingly during visualization and analysis.

All of these assumptions were made based on physical reasoning and metallurgical characteristics of brass alloys.



3.1.3 Hardness Handling

Hardness values were reported on different scales:

- Vickers Hardness (HV
- Brinell Hardness (HB)
- Rockwell B (HRB)

To prevent conversion errors, hardness was preserved as:

- Hardness_value
- Hardness_Scale

No transformations were made between scales.

3.1.4 Phase Information

Phase information, such as (α, β, α+β) and Phase volume fractions were kept as categorical variables. Because of the lack of a proper format in AI tools, they were qualitatively analyzed instead of numerical analysis.

3.2 Statistical Descriptors

The following statistical descriptors were computed on each dataset for key numerical variables:

- Mean
- Median
- Standard Deviation
- Minimum
- Maximum

These descriptors were calculated for:

- Yield Strength
- Ultimate tensile strength
- Grain size
- Elastic modulus
- Elongation

3.3 Exploratory Data Analysis

For exploring the statistical properties of the AI-generated Copper-Zinc Alloy data, Exploratory Data Analysis (EDA) was conducted. To keep the report within the drawn constraints of report length, as well as avoid redundancy in the visual presentation of the results, the analysis had to be restricted to mechanically meaningful plots.

EDA was performed separately on each of the datasets produced by ChatGPT, Gemini, Perplexity, and Copilot to maintain a neutral comparative approach.

3.3.1 Yield Strength Distribution

Histograms were used as the main method for determining the statistical distribution of mechanical strengths based on yield strength data.

The yield strength distributions indicated:

- Values are mostly within realistic ranges for Cu-Zn alloys, commonly referred to as brass, and lie between approximately 80 and approximately 450 MPa.
- Notable differences in the distribution width of AI tools.
- ChatGPT and Perplexity data showed wider distributions, which signify greater variability.
- The Gemini-generated data indicated an almost smooth distribution with moderate spreading.

- The values predicted by Copilot were more closely grouped, indicating cautious prediction.

The randomness observed within the spread of the yield strength values also suggests that neither dataset was created based on any predefined templates, but rather follows a probabilistic approach, characteristic of large language models. This randomness, while somewhat mimicking experimental randomness, occurs within a physical context.

3.3.2 Grain Size vs Yield Strength (Hall-Petch Trend)

To determine whether essential metallurgical principles are implied, yield strength is plotted against the inverse square root of grain size, as given by the Hall-Petch relation:

unpublished

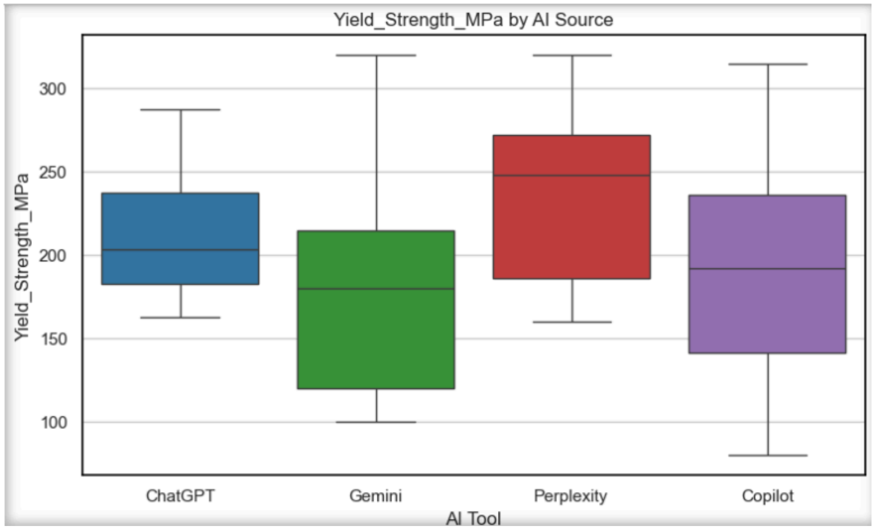
The resulting plots showed that:

$$\sigma y = \sigma 0 + kd - 1/2$$

The resulting plots demonstrated:

- A general trend for yield strength to increase with decreasing grain sizes, consistent with grain boundary strengthening mechanisms.
- Noticeable scatter around the trend line, especially for ChatGPT and Perplexity datasets. Partial deviation from linearity, where the Hall-Petch relationship was not quantitatively, but rather qualitatively, satisfied.
- Differing strengths of trends across various AI tools, with Gemini and Copilot registering slightly stronger trends.
- Thus, the findings of this section suggest that while LLMs may contain physical conceptual information, they may not be mathematically and mechanistically faithful. The trends are not calculated directly from the physics.

3.3.3 Visualization Strategy



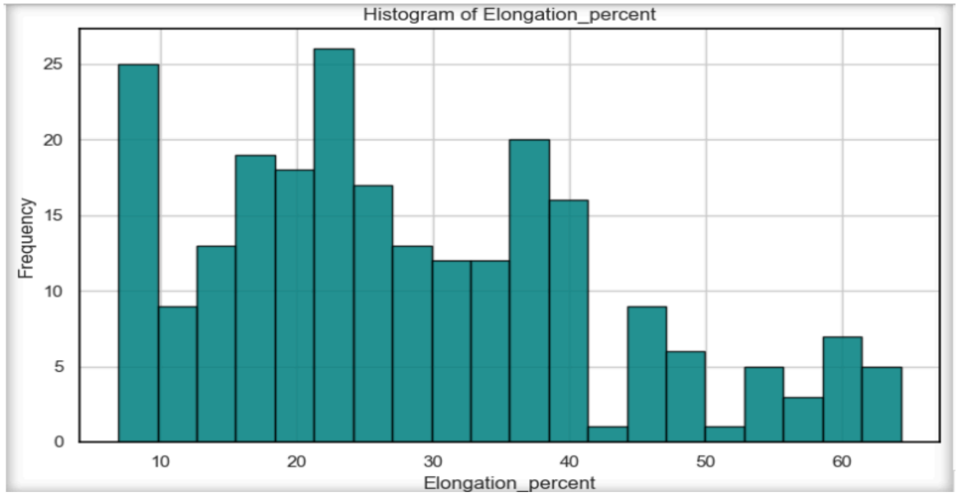
In order to comply with the page limit and minimize redundancy in visualization, a visualization strategy involving a subplot has been adopted.

Key features associated with this approach were:

- Plotting multiple, related plots (e.g., yield strength histograms provided by different AI tools) in individual figures.
- Maintaining axis limits among the subplots to enable comparison.
- The use of specific, non-yellow, non-shaded color schemes to improve readability and avoid bias.
- The use of sans-serif fonts and publication formats.

This strategy enabled a significant comparison of the data while reducing the number of figures used in the report.

3.4 Figures and Visualization Strategy



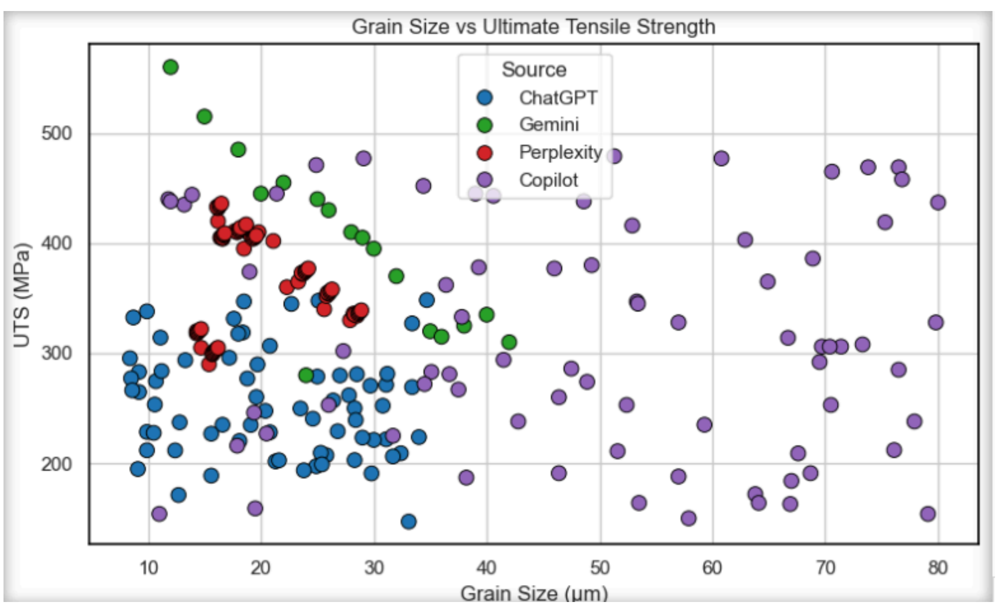
Only representative plots that are decision-critical were reported. This selection is based on their relevance and ease of interpretation.

Included visualizations were limited to:

- Yield Strength Distribution Histograms
- Grain size and yield strength relationship
- Selected distributions of processing conditions (e.g., solutionizing temperature)

All figures were produced using Python and Matplotlib and were set to high resolution to ensure publication-worthy presentation.

4. Part 3: Comparative Analysis



Once independent preprocessing and EDA have been conducted, all the AI-generated data sets are compared based on their physical plausibility, internal consistency, and conformity to the known behavior of Cu-Zn alloys.

4.1 Yield Strength Comparison Across AI Tools

A comparison of histograms and boxplots showed that there were systematic differences between AI tools:

- ChatGPT and Perplexity developed datasets that varied and had larger distributions, highlighting their more aggressive sampling approach.
- Gemini appeared to have moderate scatter with smoother trends, indicating better internal coherence.
- Conservative datasets were generated by Copilot to ensure a tighter clustering, which might be indicative of stronger implicit constraints.

The differences illustrate that there are distinct variations in how uncertainty and variability are incorporated in artificial data in AI.

4.2 Microstructure–Property Relationships

Overall qualitative metallurgical trends were noted across the datasets:

- Finer grain sizes were generally found to have higher yield strength.
- α-phase dominated alloys had higher ductility as expected for FCC structure materials.
- The presence of the β phase was correlated with an increase in strength and hardness, but a decrease in elongation.

However, the strength of the correlations varied considerably, and some datasets exhibited poor and/or non-consistent microstructure-property coupling.

4.3 Physical Plausibility and Internal Consistency

In general, most values were within physically reasonable ranges for Cu-Zn alloys. Although minor inconsistencies were found as follows:

- High elongation values at relatively high strength levels.
- Lack of strong grain size dependence in some sets, where a strong Hall-Petch effect is expected.

These imperfections approximate the type of noise one would expect to see if the experiments were truly conducted. However, they also arise due to the inherent inability of AI-generated data to ensure the mechanistic approach

4.4 Comparison with Literature Trends

When compared with the literature on Cu–Zn alloy:

- AI-generated datasets are successful in capturing qualitative trends.
- Quantitative accuracy was lacking, especially regarding consistencies in slope and magnitude.
- Scatter levels were consistently above well-controlled experimental data sets.

This comparison also illustrates the exploratory nature of AI-generated materials data.

5. Limitations of AI-Generated Materials Data

Key limitations identified:

- Lack of thermodynamic and kinetic constraints
- Inconsistent enforcement of physical laws
- Schema and unit inconsistencies
- "An occasional over-smoothing or noisy appearance."

Thus, materials data created through AI methods should be regarded as exploratory/educational, rather than predictive or even design-oriented.

6. Conclusions

This research proved the capability of large language models to synthesize artificial datasets similar to experimental materials datasets. Major metallurgical effects such as grain size strengthening and phase effects are achieved.

However, considerable scatter, internal inconsistencies, and lack of strict physical grounding are once again seen. Datasets created using AI are most useful for teaching, workflow development, and exploratory analysis, rather than materials design or engineering.

7. Deliverables Submitted

- Four AI-generated datasets (ChatGPT, Gemini, Perplexity
- Jupyter Notebook with preprocessing, EDA, and visualization codes included
- This report is about prompts, methodology, analysis, and conclusion